Simple Search Engine Report

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1 Methodology

The search engine was implemented using pipeline approach:

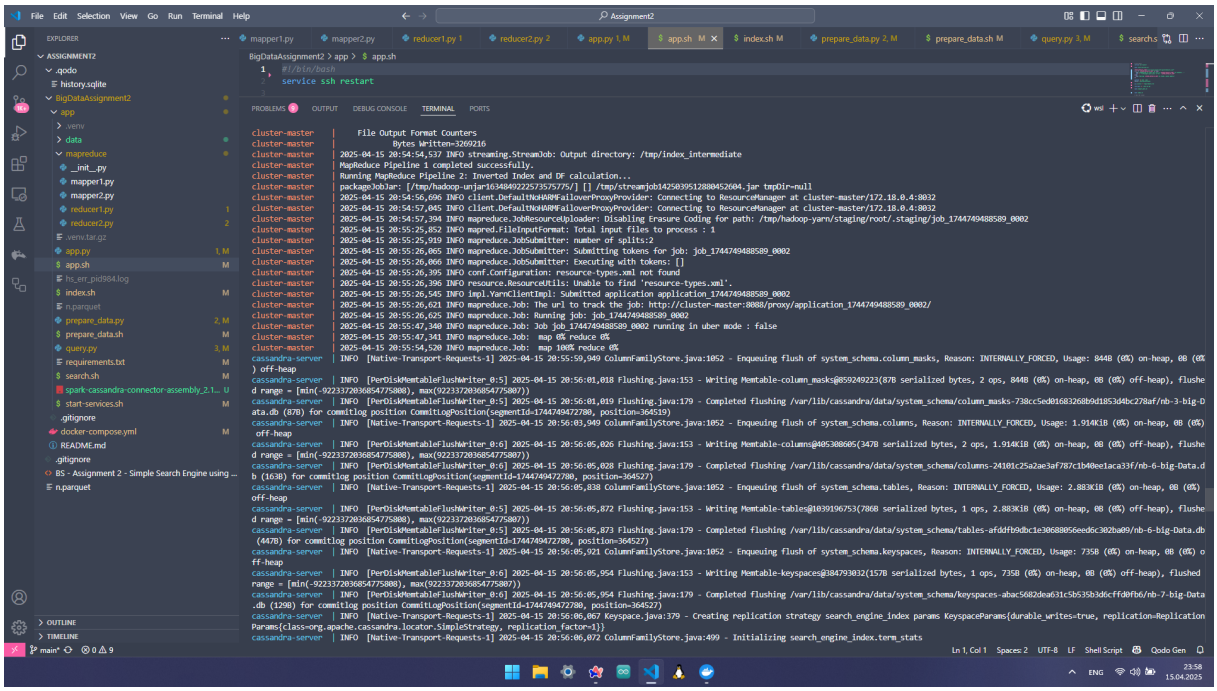
- **Data Preparation:** PySpark processed the input Parquet file (`prepare_data.py`), with relevant data extraction and creating HDFS input (`/index/data`) for the indexer.
- **Indexing:** A two-stage Hadoop MapReduce pipeline (`mapperX.py`, `reducerX.py`, managed by `index.sh`) built an inverted index and calculated document statistics (TF, DF, doc length), storing results in Cassandra tables (`doc_stats`, `term_stats`, `term_index`).
- **Ranking:** A PySpark application (`query.py`, run by `search.sh`) retrieved index data from Cassandra, calculated BM25 scores for user queries, and returned the top 10 ranked documents.

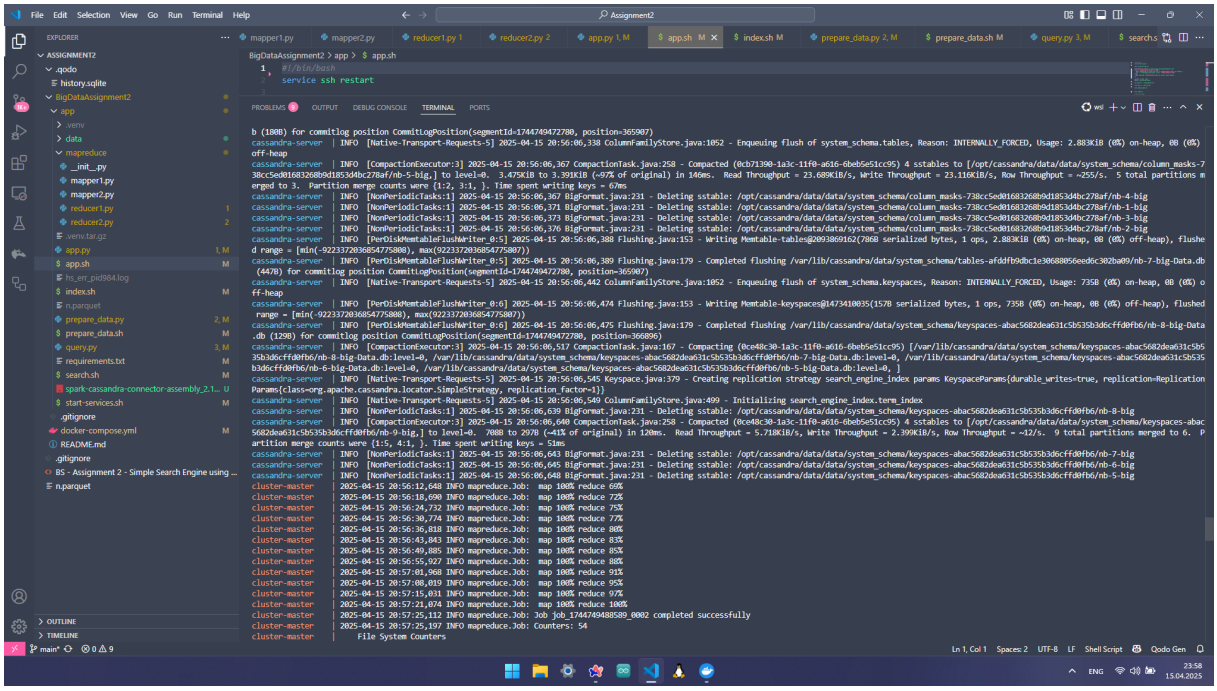
The BM25 parameters were set to $k_1 = 1.5$ and $b = 0.75$.

2 Demonstration

2.1 How to Run

1. Place `n.parquet` in the project root directory.
2. Navigate to the `BigDataAssignment2` directory.
3. Run `docker-compose up`. This builds, starts containers, prepares data, indexes, and runs a default search.
4. To run custom queries, use `docker exec -it cluster-master bash`, `cd /app`, then `./search.sh "your query"`.


File Edit Selection View Go Run Terminal Help
Assignment2
EXPLORER
ASSIGNMENT2
BigDataAssignment2
app
mapreduce
mapper1.py
mapper2.py
reducer1.py
reducer2.py
app.py
app.sh
index.sh
prepare_data.py
prepare_data.sh
query.py
requirements.txt
search.sh
start-services.sh
docker-compose.yml
README.md
.gitignore
n.parquet
BigDataAssignment2 > app > app.sh
#!/bin/bash
service ssh restart
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
cluster-master File Output Format Counters
cluster-master Bytes Written=3269216
cluster-master 2025-04-15 20:54:54,537 INFO streaming.StreamJob: Output directory: /tmp/index_intermediate
cluster-master MapReduce Pipeline 1 completed successfully.
cluster-master Running MapReduce Pipeline 2: Inverted Index and DF calculation...
cluster-master 2025-04-15 20:55:26,395 INFO conf.Configuration: resource-types.xml not found
cluster-master 2025-04-15 20:55:26,396 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
cluster-master 2025-04-15 20:55:26,545 INFO impl.YarnClientImpl: Submitted application application_1744749488589_0002
cluster-master 2025-04-15 20:55:26,621 INFO mapreduce.Job: The url to track the job: http://cluster-master:8088/proxy/application_1744749488589_0002/
cluster-master 2025-04-15 20:55:26,625 INFO mapreduce.Job: Running job: job_1744749488589_0002
cluster-master 2025-04-15 20:55:47,340 INFO mapreduce.Job: Job job_1744749488589_0002 running in uber mode : false
cluster-master 2025-04-15 20:55:47,341 INFO mapreduce.Job: map 0% reduce 0%
cluster-master 2025-04-15 20:55:54,520 INFO mapreduce.Job: map 100% reduce 0%
cassandra-server INFO [Native-Transport-Requests-1] 2025-04-15 20:55:59,949 ColumnFamilyStore.java:1052 - Enqueuing flush of system_schema.column_masks, Reason: INTERNALLY_FORCED, Usage: 844B (0%) on-heap, 0B (0%) off-heap
cassandra-server INFO [PerDiskMemtableFlushWriter_0:5] 2025-04-15 20:56:01,018 Flushing.java:153 - Writing Memtable-column_masks@859249223(87B serialized bytes, 2 ops, 844B (0%) on-heap, 0B (0%) off-heap), flushed range = [min(-9223372036854775808), max(9223372036854775807)])
cassandra-server INFO [PerDiskMemtableFlushWriter_0:5] 2025-04-15 20:56:01,019 Flushing.java:179 - Completed flushing /var/lib/cassandra/data/system_schema/column_masks-738cc5ed01683268b9d1853d4bc278af/nb-3-big-Data.db (87B) for commitlog position CommitlogPosition(segmentId=1744749472780, position=364519)
cassandra-server INFO [Native-Transport-Requests-1] 2025-04-15 20:56:03,949 ColumnFamilyStore.java:1052 - Enqueuing flush of system_schema.columns, Reason: INTERNALLY_FORCED, Usage: 1.914KiB (0%) on-heap, 0B (0%) off-heap
cassandra-server INFO [PerDiskMemtableFlushWriter_0:6] 2025-04-15 20:56:05,026 Flushing.java:153 - Writing Memtable-columns@405308605(3478 serialized bytes, 2 ops, 1.914KiB (0%) on-heap, 0B (0%) off-heap), flushed range = [min(-9223372036854775808), max(9223372036854775807)])
cassandra-server INFO [PerDiskMemtableFlushWriter_0:6] 2025-04-15 20:56:05,028 Flushing.java:179 - Completed flushing /var/lib/cassandra/data/system_schema/columns-24101c25a2ae3af787c1b40ee1aca33f/nb-6-big-Data.db (1638) for commitlog position CommitlogPosition(segmentId=1744749472780, position=364527)
cassandra-server INFO [Native-Transport-Requests-1] 2025-04-15 20:56:05,838 ColumnFamilyStore.java:1052 - Enqueuing flush of system_schema.tables, Reason: INTERNALLY_FORCED, Usage: 2.883KiB (0%) on-heap, 0B (0%) off-heap
cassandra-server INFO [PerDiskMemtableFlushWriter_0:5] 2025-04-15 20:56:05,872 Flushing.java:153 - Writing Memtable-tables@1039196753(7868 serialized bytes, 1 ops, 2.883KiB (0%) on-heap, 0B (0%) off-heap), flushed range = [min(-9223372036854775808), max(9223372036854775807)])
cassandra-server INFO [PerDiskMemtableFlushWriter_0:5] 2025-04-15 20:56:05,873 Flushing.java:179 - Completed flushing /var/lib/cassandra/data/system_schema/tables-afddfb9dbc1e30688056eed6c302ba09/nb-6-big-Data.db (4478) for commitlog position CommitlogPosition(segmentId=1744749472780, position=364527)
cassandra-server INFO [Native-Transport-Requests-1] 2025-04-15 20:56:05,921 ColumnFamilyStore.java:1052 - Enqueuing flush of system_schema.keyspaces, Reason: INTERNALLY_FORCED, Usage: 735B (0%) on-heap, 0B (0%) off-heap
cassandra-server INFO [PerDiskMemtableFlushWriter_0:6] 2025-04-15 20:56:05,954 Flushing.java:153 - Writing Memtable-keyspaces@384793032(157B serialized bytes, 1 ops, 735B (0%) on-heap, 0B (0%) off-heap), flushed range = [min(-9223372036854775808), max(9223372036854775807)])
cassandra-server INFO [PerDiskMemtableFlushWriter_0:6] 2025-04-15 20:56:05,954 Flushing.java:179 - Completed flushing /var/lib/cassandra/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-7-big-Data.db (129B) for commitlog position CommitlogPosition(segmentId=1744749472780, position=364527)
cassandra-server INFO [Native-Transport-Requests-1] 2025-04-15 20:56:05,057 Keyspace.java:379 - Creating replication strategy search_engine_index params KeyspaceParams{durable_writes=true, replication=ReplicationParams{class=org.apache.cassandra.locator.SimpleStrategy, replication_factor=1}}
cassandra-server INFO [Native-Transport-Requests-1] 2025-04-15 20:56:06,072 ColumnFamilyStore.java:499 - Initializing search_engine_index.term_stats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BigDataAssignment2
app
mapreduce
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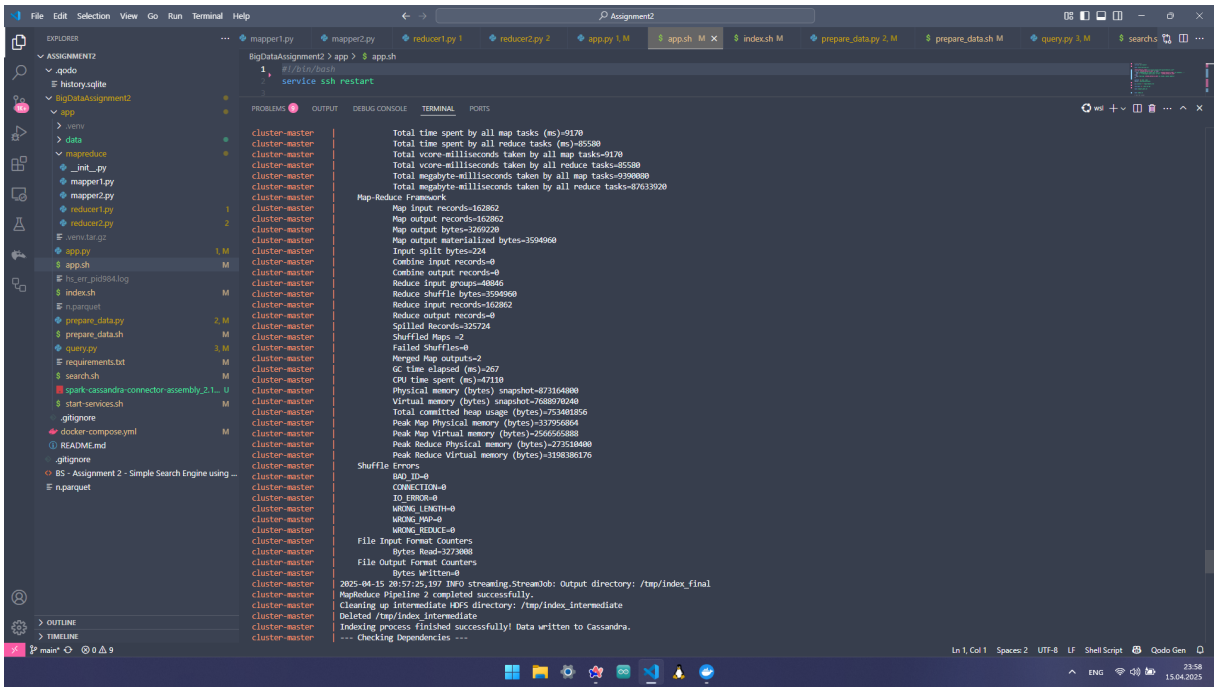
query.py
requirements.txt
search.sh
start-services.sh
docker-compose.yml
README.md
.gitignore
n.parquet
BigDataAssignment2 > app > app.sh
#!/bin/bash
service ssh restart
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
cassandra-server INFO [Native-Transport-Requests-5] 2025-04-15 20:56:06,338 ColumnFamilyStore.java:1052 - Enqueuing flush of system_schema.tables, Reason: INTERNALLY_FORCED, Usage: 2.883KiB (0%) on-heap, 0B (0%) off-heap
cassandra-server INFO [CompactionExecutor:3] 2025-04-15 20:56:06,367 CompactionTask.java:258 - Compacted (0cb71300-1a3c-11f0-a616-6beb5e51cc95) 4 sstables to [/opt/cassandra/data/data/system_schema/column_masks-738cc5ed01683268b9d1853d4bc278af/nb-5-big,] to level=0. 3.475KiB to 3.399KiB (~97% of original) in 140ms. Read Throughput = 23.608KiB/s, Write Throughput = 23.110KiB/s, Row Throughput = ~255/s. 5 total partitions merged to 3. Partition merge counts were {1:2, 3:1, }. Time spent writing keys = 67ms
cassandra-server INFO [NonPeriodicTasks:1] 2025-04-15 20:56:06,367 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/column_masks-738cc5ed01683268b9d1853d4bc278af/nb-4-big
cassandra-server INFO [NonPeriodicTasks:1] 2025-04-15 20:56:06,371 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/column_masks-738cc5ed01683268b9d1853d4bc278af/nb-1-big
cassandra-server INFO [NonPeriodicTasks:1] 2025-04-15 20:56:06,373 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/column_masks-738cc5ed01683268b9d1853d4bc278af/nb-3-big
cassandra-server INFO [NonPeriodicTasks:1] 2025-04-15 20:56:06,376 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/column_masks-738cc5ed01683268b9d1853d4bc278af/nb-2-big
cassandra-server INFO [PerDiskMemtableFlushWriter_0:5] 2025-04-15 20:56:06,388 Flushing.java:153 - Writing Memtable-tables@2093869162(7868 serialized bytes, 1 ops, 2.883KiB (0%) on-heap, 0B (0%) off-heap), flushed range = [min(-9223372036854775808), max(9223372036854775807)])
cassandra-server INFO [PerDiskMemtableFlushWriter_0:5] 2025-04-15 20:56:06,389 Flushing.java:179 - Completed flushing /var/lib/cassandra/data/system_schema/tables-afddfb9dbc1e30688056eed6c302ba09/nb-7-big-Data.db (4478) for commitlog position CommitlogPosition(segmentId=1744749472780, position=365907)
cassandra-server INFO [Native-Transport-Requests-5] 2025-04-15 20:56:06,442 ColumnFamilyStore.java:1052 - Enqueuing flush of system_schema.keyspaces, Reason: INTERNALLY_FORCED, Usage: 735B (0%) on-heap, 0B (0%) off-heap
cassandra-server INFO [PerDiskMemtableFlushWriter_0:6] 2025-04-15 20:56:06,474 Flushing.java:153 - Writing Memtable-keyspaces@1473410035(157B serialized bytes, 1 ops, 735B (0%) on-heap, 0B (0%) off-heap), flushed range = [min(-9223372036854775808), max(9223372036854775807)])
cassandra-server INFO [PerDiskMemtableFlushWriter_0:6] 2025-04-15 20:56:06,475 Flushing.java:179 - Completed flushing /var/lib/cassandra/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-8-big-Data.db (129B) for commitlog position CommitlogPosition(segmentId=1744749472780, position=366896)
cassandra-server INFO [CompactionExecutor:3] 2025-04-15 20:56:06,517 CompactionTask.java:167 - Compacting (0ce48c30-1a3c-11f0-a616-6beb5e51cc95) [/var/lib/cassandra/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-8-big-Data.db:level=0, /var/lib/cassandra/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-7-big-Data.db:level=0, /var/lib/cassandra/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-6-big-Data.db:level=0, /var/lib/cassandra/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-5-big-Data.db:level=0,]
cassandra-server INFO [Native-Transport-Requests-5] 2025-04-15 20:56:06,545 Keyspace.java:379 - Creating replication strategy search_engine_index params KeyspaceParams{durable_writes=true, replication=ReplicationParams{class=org.apache.cassandra.locator.SimpleStrategy, replication_factor=1}}
cassandra-server INFO [Native-Transport-Requests-5] 2025-04-15 20:56:06,549 ColumnFamilyStore.java:499 - Initializing search_engine_index.term_index
cassandra-server INFO [NonPeriodicTasks:1] 2025-04-15 20:56:06,639 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-8-big
cassandra-server INFO [CompactionExecutor:3] 2025-04-15 20:56:06,640 CompactionTask.java:258 - Compacted (0ce48c30-1a3c-11f0-a616-6beb5e51cc95) 4 sstables to [/opt/cassandra/data/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-9-big,] to level=0. 7088 to 2978 (~41% of original) in 120ms. Read Throughput = 5.718KiB/s, Write Throughput = 2.399KiB/s, Row Throughput = ~12/s. 9 total partitions merged to 6. Partition merge counts were {1:5, 4:1, }. Time spent writing keys = 51ms
cassandra-server INFO [NonPeriodicTasks:1] 2025-04-15 20:56:06,643 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-7-big
cassandra-server INFO [NonPeriodicTasks:1] 2025-04-15 20:56:06,645 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-6-big
cassandra-server INFO [NonPeriodicTasks:1] 2025-04-15 20:56:06,648 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/keyspaces-abac5682dea631c5b535b3d6cffd0fb6/nb-5-big
cluster-master 2025-04-15 20:56:12,648 INFO mapreduce.Job: map 100% reduce 69%
cluster-master 2025-04-15 20:56:18,690 INFO mapreduce.Job: map 100% reduce 72%
cluster-master 2025-04-15 20:56:24,732 INFO mapreduce.Job: map 100% reduce 75%
cluster-master 2025-04-15 20:56:30,774 INFO mapreduce.Job: map 100% reduce 77%
cluster-master 2025-04-15 20:56:36,818 INFO mapreduce.Job: map 100% reduce 80%
cluster-master 2025-04-15 20:56:43,843 INFO mapreduce.Job: map 100% reduce 83%
cluster-master 2025-04-15 20:56:49,885 INFO mapreduce.Job: map 100% reduce 85%
cluster-master 2025-04-15 20:56:55,927 INFO mapreduce.Job: map 100% reduce 88%
cluster-master 2025-04-15 20:57:01,968 INFO mapreduce.Job: map 100% reduce 91%
cluster-master 2025-04-15 20:57:08,019 INFO mapreduce.Job: map 100% reduce 93%
cluster-master 2025-04-15 20:57:15,031 INFO mapreduce.Job: map 100% reduce 97%
cluster-master 2025-04-15 20:57:21,074 INFO mapreduce.Job: map 100% reduce 100%
cluster-master 2025-04-15 20:57:25,112 INFO mapreduce.Job: Job job_1744749488589_0002 completed successfully
cluster-master 2025-04-15 20:57:25,197 INFO mapreduce.Job: Counters: 54
cluster-master File System Counters


File Edit Selection View Go Run Terminal Help
Assignment2
EXPLORER
ASSIGNMENT2
.qodo
history.sqlite
BigDataAssignment2
app
.venv
data
mapreduce
__init__.py
mapper1.py
mapper2.py
reducer1.py
reducer2.py
.venv.tar.gz
app.py
app.sh
hs_err_pid984.log
index.sh
n.parquet
prepare_data.py
prepare_data.sh
query.py
requirements.txt
search.sh
spark-cassandra-connector-assembly_2.1...
start-services.sh
.gitignore
docker-compose.yml
README.md
.gitignore
BS - Assignment 2 - Simple Search Engine using ...
n.parquet
OUTLINE
TIMELINE
mapper1.py
mapper2.py
reducer1.py 1
reducer2.py 2
app.py 1, M
app.sh M
index.sh M
prepare_data.py 2, M
prepare_data.sh M
query.py 3, M
BigDataAssignment2 > app > app.sh
#!/bin/bash
service ssh restart
PROBLEMS 7
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
wsl
cluster-master | Total time spent by all map tasks (ms)=9170
cluster-master | Total time spent by all reduce tasks (ms)=85580
cluster-master | Total vcore-milliseconds taken by all map tasks=9170
cluster-master | Total vcore-milliseconds taken by all reduce tasks=85580
cluster-master | Total megabyte-milliseconds taken by all map tasks=9390080
cluster-master | Total megabyte-milliseconds taken by all reduce tasks=87633920
cluster-master | Map-Reduce Framework
cluster-master | Map input records=162862
cluster-master | Map output records=162862
cluster-master | Map output bytes=3269220
cluster-master | Map output materialized bytes=3594960
cluster-master | Input split bytes=224
cluster-master | Combine input records=0
cluster-master | Combine output records=0
cluster-master | Reduce input groups=40846
cluster-master | Reduce shuffle bytes=3594960
cluster-master | Reduce input records=162862
cluster-master | Reduce output records=0
cluster-master | Spilled Records=325724
cluster-master | Shuffled Maps =2
cluster-master | Failed Shuffles=0
cluster-master | Merged Map outputs=2
cluster-master | GC time elapsed (ms)=267
cluster-master | CPU time spent (ms)=47110
cluster-master | Physical memory (bytes) snapshot=873164800
cluster-master | Virtual memory (bytes) snapshot=7688970240
cluster-master | Total committed heap usage (bytes)=753401856
cluster-master | Peak Map Physical memory (bytes)=337956864
cluster-master | Peak Map Virtual memory (bytes)=2566565888
cluster-master | Peak Reduce Physical memory (bytes)=273510400
cluster-master | Peak Reduce Virtual memory (bytes)=3198386176
cluster-master | Shuffle Errors
cluster-master | BAD_ID=0
cluster-master | CONNECTION=0
cluster-master | IO_ERROR=0
cluster-master | WRONG_LENGTH=0
cluster-master | WRONG_MAP=0
cluster-master | WRONG_REDUCE=0
cluster-master | File Input Format Counters
cluster-master | Bytes Read=1273808
cluster-master | File Output Format Counters
cluster-master | Bytes Written=0
cluster-master | 2025-04-15 20:57:25,197 INFO streaming.StreamJob: Output directory: /tmp/index_final
cluster-master | MapReduce Pipeline 2 completed successfully.
cluster-master | Cleaning up intermediate HDFS directory: /tmp/index_intermediate
cluster-master | Deleted /tmp/index_intermediate
cluster-master | Indexing process finished successfully! Data written to Cassandra.
cluster-master | --- Checking Dependencies ---
main*
Ln 1, Col 1
Spaces: 2
UTF-8
LF
Shell Script
Qodo Gen
ENG
23:58
15.04.2025